HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 6: Operational Amplifiers

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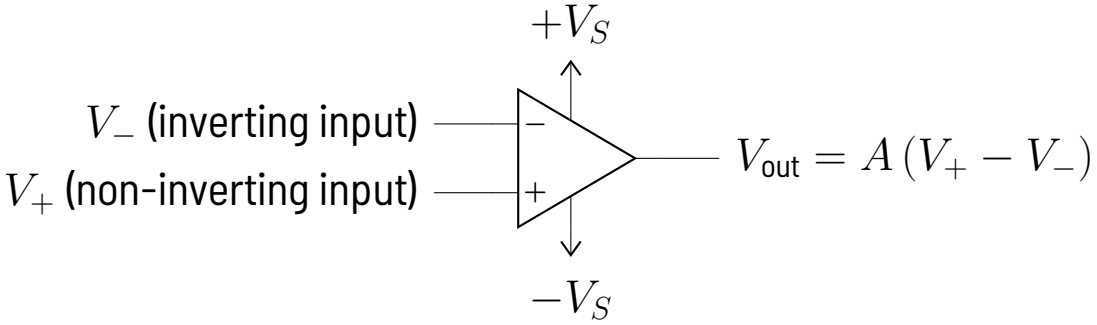
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6 Operational Amplifiers

6.1 The Ideal Operational Amplifier

The op-amp: symbol and model

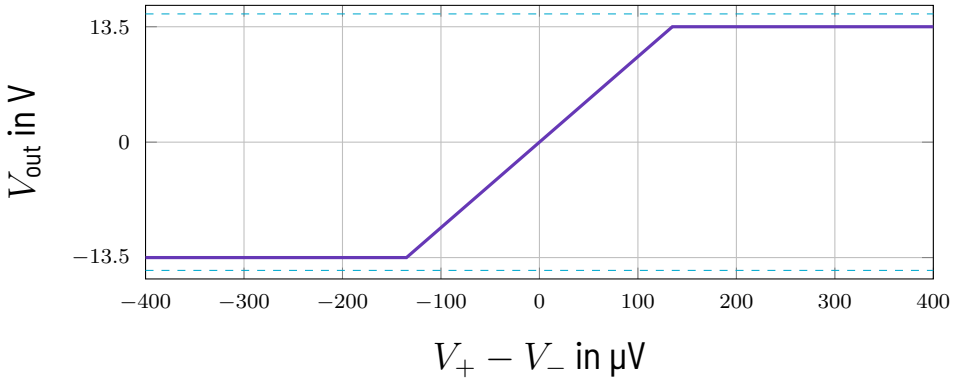


- amplifies the **difference** of its two input voltages
- open-loop gain A: typically $10^5 \ldots 10^7$ (100 dB ... 140 dB)
- supply $\pm V_S$ (or single supply V_S and ground); the output cannot leave the supply range
- supply pins are often omitted in drawings – but they are always there

Properties of the ideal op-amp

property	ideal	real (typical)
open-loop gain A	∞	$10^5 \ldots 10^7$
input resistance	∞	1 MΩ ...1 TΩ
input currents	0	pA (CMOS) ...100 nA (bipolar)
output resistance	0	$10\,\Omega$ ... $100\,\Omega$ (open loop)
offset voltage	0	10 µV ...5 mV
bandwidth	∞	GBW 1 MHz ...100 MHz

Open loop: the op-amp as a comparator



- with $A = 10^5$ and ± 13.5 V output swing the linear range is only ± 135 µV
- any larger difference drives the output to a rail: V_{out} tells only **which input is higher**
- this is a **comparator** – the building block of the flash ADC (chapter 9)

Negative feedback and the golden rules

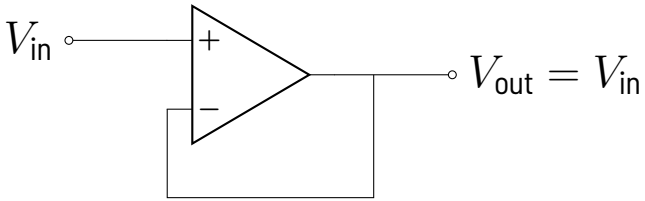
- **negative feedback**: a part of V_{out} is fed back to the **inverting** input
- if V_{out} rises, V_- rises, the difference $V_+ - V_-$ shrinks and pulls V_{out} back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$

Rule 6.1 ***Golden rules*** *for an op-amp with negative feedback:*

1. *The output does whatever is necessary to make the voltage difference between the inputs zero:* $V_+ = V_-$.
2. *The inputs draw no current:* $I_+ = I_- = 0$.

6.2 Basic Amplifier Circuits

Voltage follower (buffer)



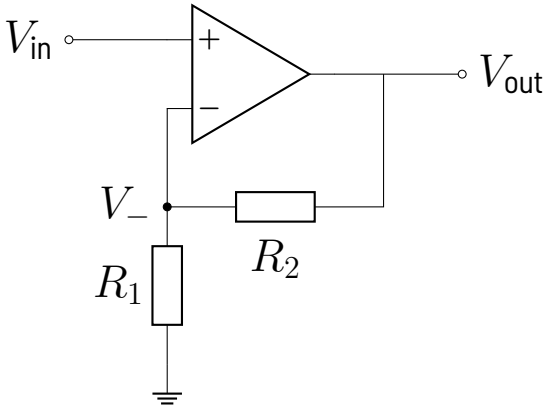
- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$, gain 1
- input current ≈ 0: the source is **not loaded**
- output: low resistance, can drive mA into a load

Why buffering matters

- divider $100\,\text{k}\Omega/100\,\text{k}\Omega$ at $10\,\text{V}$, load $R_L = 10\,\text{k}\Omega$
- directly: $V_{\text{out}} = 10\,\text{V} \cdot \frac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L} = 0.833\,\text{V}$ instead of $5\,\text{V}$
- behind a follower: $5.00\,\text{V}$, the divider does not notice the load
- uses in this course
  - bridge taps $\rightarrow$ instrumentation amplifier inputs
  - reference ladder of the flash ADC, hold capacitor of the S&H

▶ Falstad: `examples/opamps/opamp_follower.txt`

Non-inverting amplifier

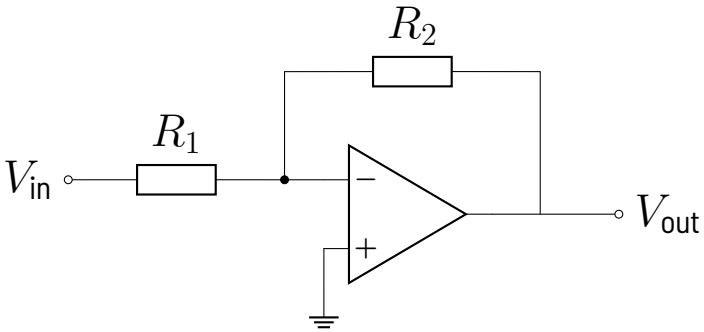


- R_1, R_2 form a divider: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$

Inverting amplifier

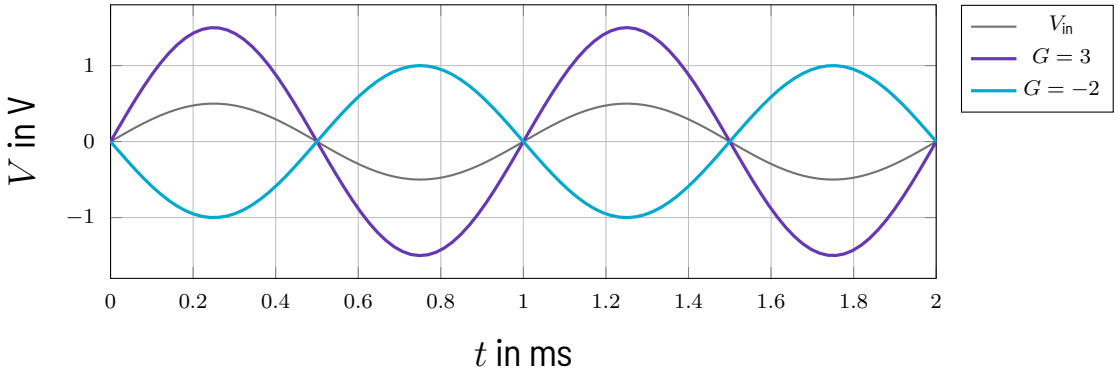


- rule 1: $V_- = V_+ = 0$, the inverting input is a **virtual ground**
- rule 2: the current V_{in}/R_1 continues through R_2: $V_{\text{out}} = -R_2 \cdot V_{\text{in}}/R_1$
- input resistance $= R_1$ (it loads the source!)

Inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1}$$

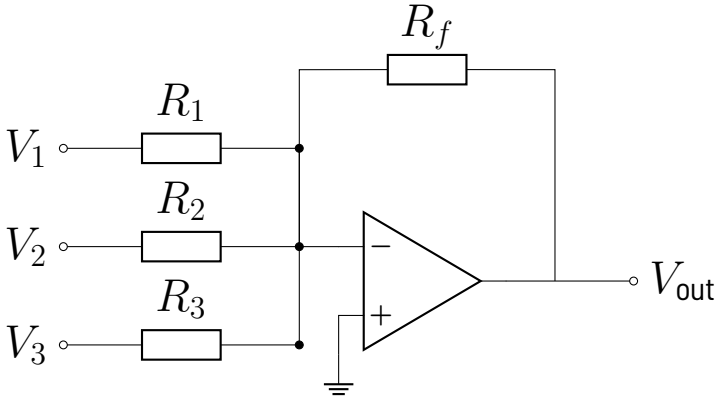
Non-inverting and inverting in Falstad



- V_{in}: 0.5 V amplitude, 1 kHz; $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ in both circuits
- non-inverting: $G = 1 + 2 = 3$, 1.5 V in phase; inverting: $G = -2$, 1.0 V, phase $180°$

▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`

Summing amplifier



- the virtual ground decouples the inputs; the currents add up in R_f

Inverting summing amplifier

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

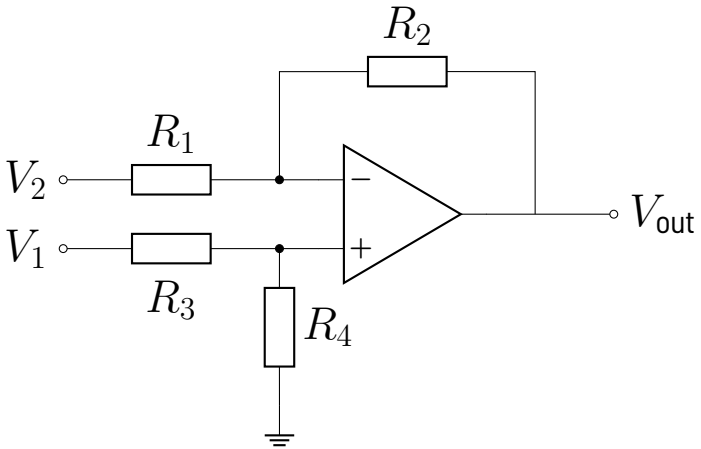
6.3 Difference and Instrumentation Amplifiers

Differential and common-mode signal

- two input voltages V_1, V_2 can be written as
  - differential-mode voltage $V_D = V_1 - V_2$ (the measurement signal)
  - common-mode voltage $V_{CM} = (V_1 + V_2)/2$ (not wanted)
- symmetric bridge at $5\,\text{V}$: V_D some mV, $V_{CM} \approx 2.5\,\text{V}$; capstone bridge: $V_D = 0\,\text{mV} \ldots 37\,\text{mV}$, $V_{CM} \approx 0.1\,\text{V}$
- ideal: $V_{\text{out}} = A_D\, V_D$; real: $V_{\text{out}} = A_D\, V_D + A_{CM}\, V_{CM}$

Definition 6.1 *The **common-mode rejection ratio** is the ratio of differential gain to common-mode gain:* $\text{CMRR} = A_D/A_{CM}$, *in decibels* $20 \log_{10}(A_D/A_{CM})$.

Difference amplifier



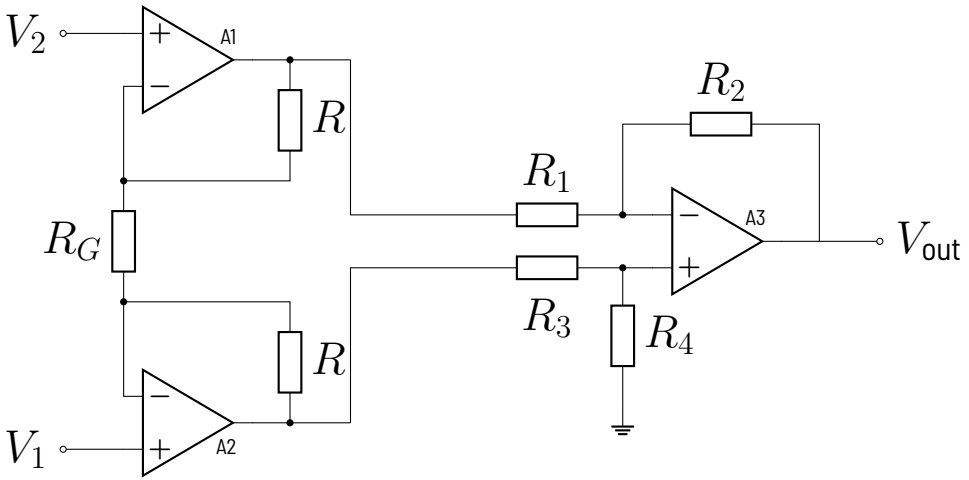
- superposition: $V_{\text{out}} = V_1 \dfrac{R_4}{R_3 + R_4}\left(1 + \dfrac{R_2}{R_1}\right) - V_2 \dfrac{R_2}{R_1}$
- with $R_4/R_3 = R_2/R_1$: $V_{\text{out}} = \dfrac{R_2}{R_1}(V_1 - V_2)$

Resistor matching limits the CMRR

- the common-mode rejection depends only on how well R_4/R_3 equals R_2/R_1
- example $G = 10$, $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$: ideal $V_\text{out} = 50.0\,\text{mV}$
- R_4 only $1\,\%$ too high: $V_\text{out} = 72.6\,\text{mV}$ – an error of $45\,\%$!
- worst case with resistor tolerance ε: CMRR $\approx \dfrac{1+G}{4\varepsilon}$
  - $1\,\%$ resistors, $G = 10$: CMRR ≈ 275 ($49\,\text{dB}$)
  - remedy: $0.1\,\%$ resistors, matched resistor networks, a trimmer, or an integrated difference amplifier (e.g. INA105, AD8276)

▶ Falstad: `examples/opamps/opamp_diffamp.txt`

Instrumentation amplifier (three op-amps)



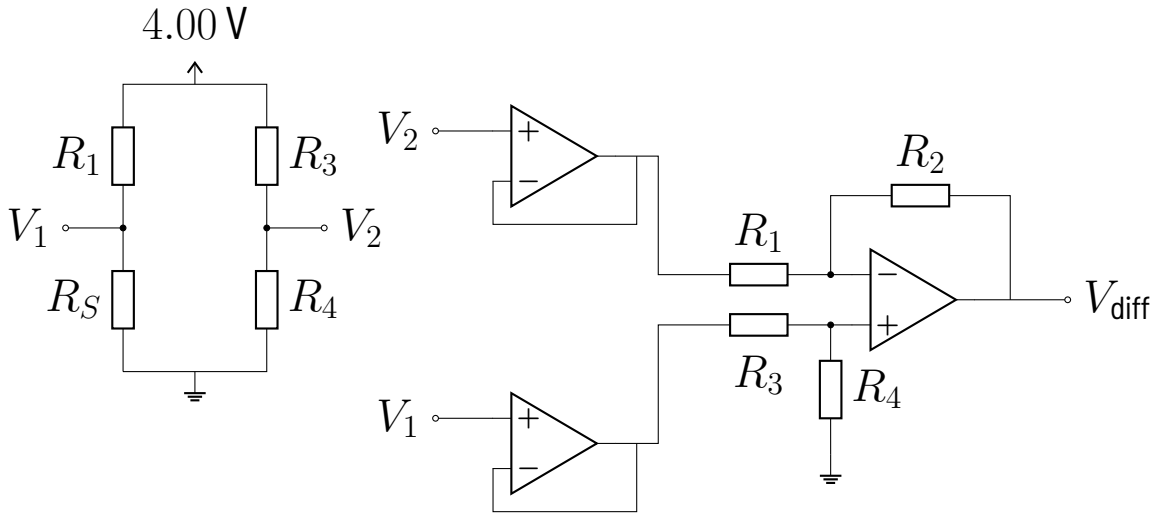
Instrumentation amplifier

$$V_{\text{out}} = \left(1 + \frac{2R}{R_G}\right) \frac{R_2}{R_1} (V_1 - V_2)$$

How the three op-amp INA works

- A1, A2 are non-inverting: the inputs draw no current, the bridge is not loaded
- rule 1 puts V_1 and V_2 across R_G: current $(V_1 - V_2)/R_G$ flows through R, R_G, R
- difference at the outputs of A1/A2: $(V_1 - V_2)(1 + 2R/R_G)$; the common mode passes with gain 1
- A3 is a difference amplifier with gain R_2/R_1 and removes the common mode
- one resistor R_G sets the gain; integrated INAs (INA128, AD620, AD8221) contain everything except R_G

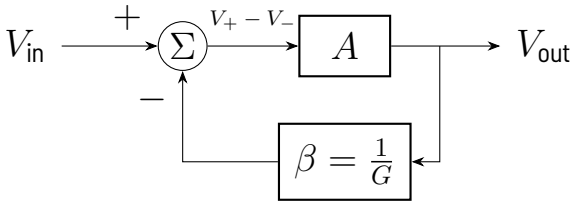
The lab version: two buffers and a difference amplifier



- followers isolate the bridge; the difference stage sets the gain $G_1 = R_2/R_1$
- simpler than the three op-amp version, but all gain must come from R_2/R_1

6.4 Limits of Real Op-Amps

Finite open-loop gain



- $V_{\text{out}} = A\,(V_{\text{in}} - \beta\,V_{\text{out}})$, with the feedback factor $\beta = 1/G$ (non-inverting: $\beta = R_1/(R_1 + R_2)$)
- loop gain $A\beta = A/G$: the larger, the closer the circuit comes to the ideal gain G

Closed-loop gain with finite open-loop gain A

$$G_{\text{real}} = \frac{A}{1 + A\beta} = \frac{G}{1 + G/A} = \frac{1}{1/G + 1/A} \qquad \text{relative error} \approx -\frac{G}{A}$$

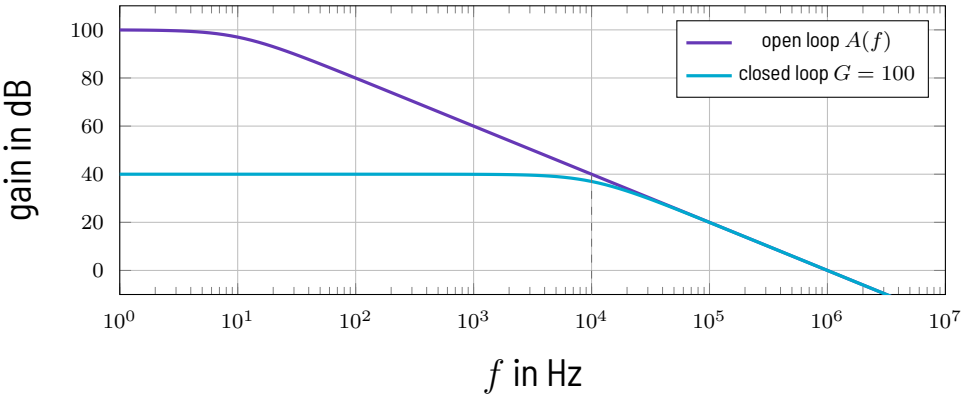
How large is the error?

ideal gain G	G_{real} ($A = 10^5$)	error	V_{out} for 1 mV
10	9.999	$-0.01\,\%$	10.0 mV
100	99.90	$-0.1\,\%$	99.9 mV
1000	990.1	$-1\,\%$	0.990 V
10000	9091	$-9.1\,\%$	9.09 V

- the error grows proportionally to G: one stage with $G > 1000$ shows a visible error
- **split the gain**: $G = 10 \cdot 100$ gives 998.9 ($-0.11\,\%$), $G = 100 \cdot 100$ gives 9980 ($-0.2\,\%$)
- rule of the lab: aim for $G < 100$ per stage

▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

Gain-bandwidth product



- above a low corner frequency (here 10 Hz) A falls by 20 dB per decade: $A(f) \approx \text{GBW}/f$
- closed-loop bandwidth $f_c \approx \text{GBW}/G$ (here 1 MHz/100 = 10 kHz)

Consequences of the GBW

- on the falling slope gain and frequency are inversely proportional:

Gain-bandwidth product

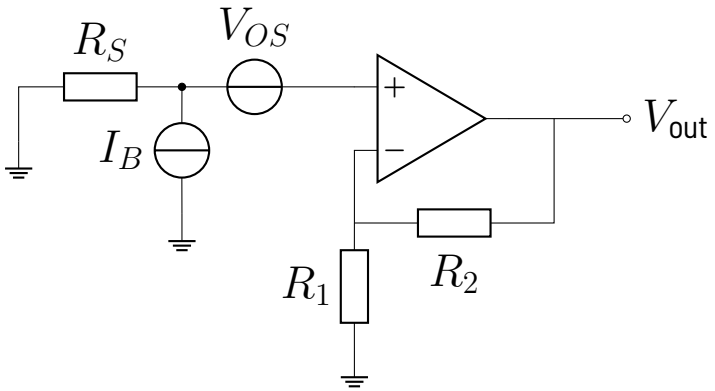
GBW $= A(f) \cdot f = G \cdot f_c$ (constant above the corner frequency)

- at 1 kHz an op-amp with GBW = 1 MHz has only $A = 1000$: a $G = 100$ stage then has $10\,\%$ error
- splitting the gain also helps here: two stages $G = 10$ each have 100 kHz bandwidth
- our sensor signal is slow (a few Hz): the GBW is no problem in the capstone
- **Falstad does not simulate the GBW**: its op-amp has a constant $A = 10^5$ at all frequencies

Output swing and rail-to-rail

- the output cannot exceed the supply; a classic op-amp stays $1\,\mathrm{V}$ … $2\,\mathrm{V}$ away from the rails
  - LM358 at $5\,\mathrm{V}$: output from about $0\,\mathrm{V}$ to $3.5\,\mathrm{V}$
- **rail-to-rail output**: reaches the rails within some $10\,\mathrm{mV}$ (light load) to a few $100\,\mathrm{mV}$ (mA load)
- the **input common-mode range** is limited as well: not every op-amp works with inputs near the rails (**rail-to-rail input**); the capstone bridge sits at only $0.1\,\mathrm{V}$
- consequence for the capstone: an output of exactly $0\,\mathrm{V}$ is not possible on a single $5\,\mathrm{V}$ supply – use a negative supply, or a small offset and a range of e.g. $0.1\,\mathrm{V}$ … $4.1\,\mathrm{V}$

Input offset voltage and bias currents



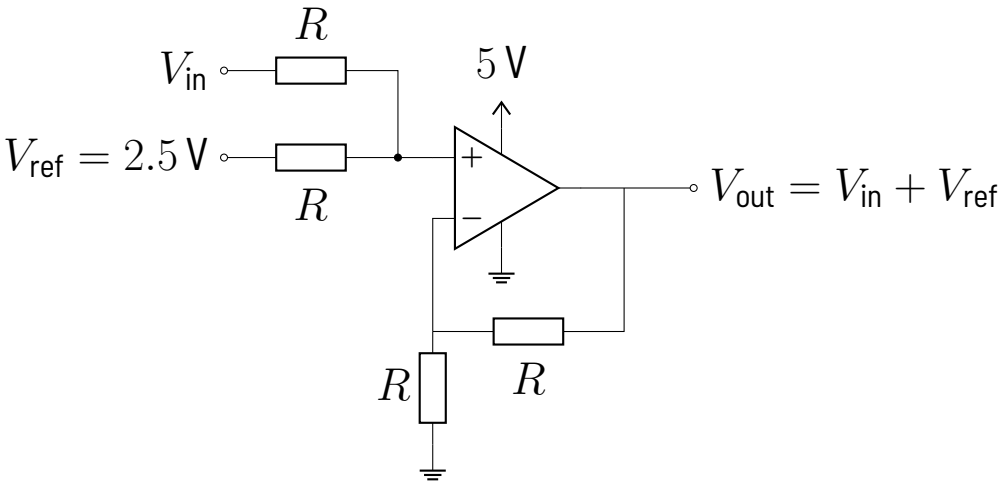
- offset V_{OS}: a small voltage source in series with an input, amplified by $1 + R_2/R_1$
- bias current I_B flows through the source resistance R_S and adds $R_S I_B$
- $V_{\text{out}} = (1 + R_2/R_1)\,(V_{OS} + R_S I_B)$ with $V_{\text{in}} = 0$

Offset after a high gain

- typical values
  - general purpose bipolar (LM358): $V_{OS} \approx 2\,\text{mV}$, $I_B \approx 20\,\text{nA}$
  - CMOS (MCP6002, TLV9002): $V_{OS} \approx 1\,\text{mV}$, $I_B \approx 1\,\text{pA}$
  - zero-drift / chopper (OPA2333, MCP6V02): $V_{OS} < 10\,\mu\text{V}$
- Falstad example with $G = 101$: $V_{OS} = 1\,\text{mV}$ gives $0.101\,\text{V}$; $I_B = 100\,\text{nA}$ through $100\,\text{k}\Omega$ gives $-1.01\,\text{V}$
- capstone: $2\,\text{mV}$ offset $\times$ 107.6 = $0.22\,\text{V}$ – $5\,\%$ of the full scale, almost half an LSB!
- remedies: zero-drift op-amps, CMOS inputs, low source resistances, **offset trimming**

▶ Falstad: `examples/opamps/opamp_offset.txt`

Single-supply operation and level shifting

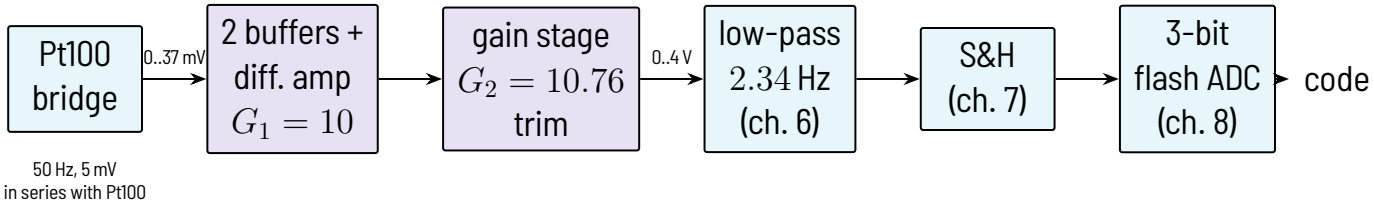


- single supply: the output can only be positive; bipolar signals need a **bias point** (often $V_S/2$)
- here $V_+ = (V_{\text{in}} + V_{\text{ref}})/2$, gain 2: $V_{\text{in}} = \pm 2\,\text{V}$ becomes $0.5\,\text{V} \ldots 4.5\,\text{V}$
- a follower on the same supply clips the negative half wave at $0\,\text{V}$

▶ Falstad: `examples/opamps/opamp_level_shift.txt`

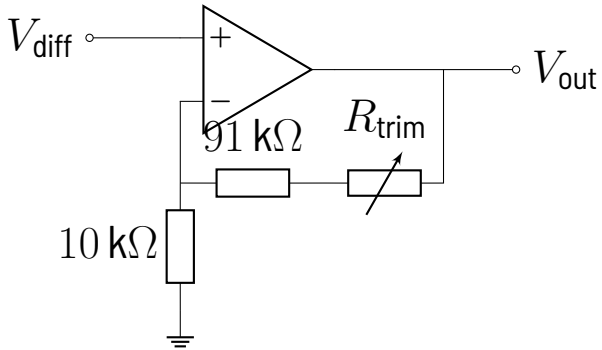
6.5 Designing the Measurement Amplifier

The signal chain of the capstone



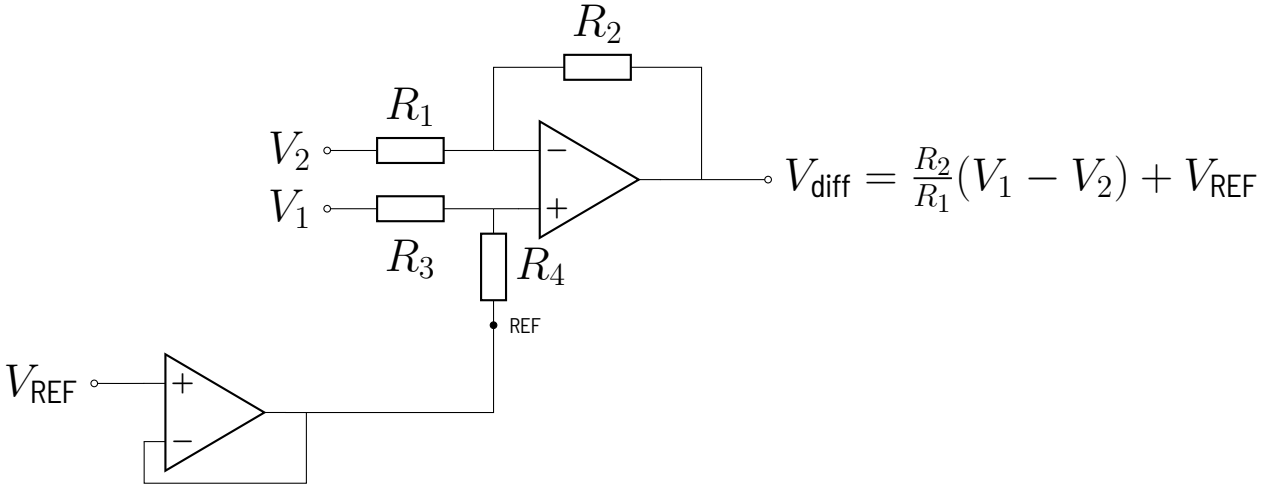
- bridge from $4.00\,\text{V}$: $V_{BR} = V_{\text{ref}} \left(\frac{R}{R_1+R} - \frac{R_4}{R_3+R_4} \right) = 0\,\text{mV} \ldots 37.18\,\text{mV}$ for $R = 100\,\Omega \ldots 138.5\,\Omega$
- total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: difference amplifier $G_1 = 100\,\text{k}\Omega/10\,\text{k}\Omega = 10$, non-inverting stage $G_2 = 10.76$ – both < 100
- zero: set by the bridge ($R_4 = R(0\,°\text{C})$); span: one resistor R_F in the second stage

Gain stage with trimmer



- required: $G_2 = 4.00\,\text{V}/(10 \cdot 37.18\,\text{mV}) = 10.758 \Rightarrow R_F = 97.58\,\text{k}\Omega$
- fixed E96 value: $R_F = 97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$ (used in Falstad)
- in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer: $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V} \ldots 4.13\,\text{V}$; setting $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- the fixed resistor carries most of the value: fine trimmer resolution, enough range for the tolerances

Zero adjustment at the reference input



- the lower end of R_4 is the **reference input**: V_{REF} adds directly to the output
- it must be driven from a low impedance (follower), otherwise it changes R_4 and the CMRR
- in the capstone the zero is set by the bridge ($R_4 = R(0\,^\circ\text{C})$); a REF adjustment of a few mV is needed only if op-amp offsets or resistor tolerances shift it

Trimming procedure

- two adjustments: zero (offset) first, then span (gain)

Design card: Measurement amplifier (zero and span)

1. Calculate the bridge span $V_{BR,\text{max}}$ and the total gain $G = V_{\text{FS}}/V_{BR,\text{max}}$ (here $4.00\,\text{V}/37.18\,\text{mV} = 107.6$).
2. Split the gain: input stage (buffers + difference amplifier) $G_1 = 10$, second stage $G_2 = G/G_1 = 10.76$, each stage $G < 100$.
3. Use a fixed resistor for most of R_F and a trimmer for the rest ($91\,\text{k}\Omega$ + $10\,\text{k}\Omega$).
4. **Zero**: set the sensor to the bottom of the range ($100\,\Omega$, or a precise resistor) and check $V_{\text{out}} = 0.00\,\text{V}$; correct with R_4 or V_{REF} if necessary.
5. **Span**: set the sensor to the top of the range ($138.5\,\Omega$), adjust R_{trim} until $V_{\text{out}} = 4.00\,\text{V}$.
6. Check the zero again.

The complete front end in Falstad

- bridge 4.00 V, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100 with a slider from $100\,\Omega$ to $138.5\,\Omega$
- two followers, difference amplifier $10\,\text{k}\Omega/100\,\text{k}\Omega$, gain stage $1 + (91\,\text{k}\Omega + R_{\text{trim}})/10\,\text{k}\Omega$, $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- readings: $V_R = 0.100$ V, $V_{BR} = 37.18$ mV, $V_{\text{diff}} = 372$ mV, $V_{\text{out}} = 4.00$ V at $138.5\,\Omega$; $V_{\text{out}} = 0.00$ V at $100\,\Omega$
- the 50 Hz interference of 5 mV in series with the Pt100 is a **differential** signal: 4.9 mV at the tap, $\times 107.6 = 0.52$ V amplitude
  - no CMRR helps here – the low-pass filter of chapter 7 has to remove it
  - the output must have headroom: $4\,\text{V} + 0.52\,\text{V}$ peak

▶ Falstad: `examples/opamps/opamp_ina.txt`

6.6 Exercises

Exercise 6.1: Non-inverting and inverting amplifier

A sine voltage with 0.5 V amplitude and 1 kHz is to be amplified. Resistors: $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$.

a) Calculate gain and output amplitude of the non-inverting amplifier.

b) Calculate gain and output amplitude of the inverting amplifier. What is the phase shift?

c) What input resistance does the source see in both circuits?

d) Which resistor would you change to get an inverting gain of -4.7?

e) Check a) and b) in Falstad.

Solution 6.1

a) $G = 1 + R_2/R_1 = 1 + 20/10 = 3$; output amplitude $3 \cdot 0.5\,\text{V} = 1.5\,\text{V}$, in phase with the input.

b) $G = -R_2/R_1 = -2$; output amplitude $1.0\,\text{V}$, phase shift $180°$.

c) Non-inverting: practically infinite (the op-amp input, MΩ to TΩ). Inverting: $R_1 = 10\,\text{k}\Omega$, because the inverting input is a virtual ground.

d) $R_2 = 4.7 \cdot R_1 = 47\,\text{k}\Omega$ (E12 value).

e) ▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`
the scope shows `Vout_ni` with $\pm 1.50\,\text{V}$ and `Vout_inv` with $\pm 1.00\,\text{V}$, inverted.

Exercise 6.2: Summing amplifier

An inverting summing amplifier has $R_f = 10\,\text{k}\Omega$ and three inputs: $V_1 = 1\,\text{V}$ via $R_1 = 10\,\text{k}\Omega$, $V_2 = 2\,\text{V}$ via $R_2 = 20\,\text{k}\Omega$, $V_3 = -0.5\,\text{V}$ via $R_3 = 10\,\text{k}\Omega$.

a) Calculate V_{out} and the voltage at the inverting input.
b) Calculate the current through R_f.
c) How must R_3 be changed so that V_3 is weighted with -0.25?
d) Check a) in Falstad.

Solution 6.2

a) $V_{\text{out}} = -10\,\text{k}\Omega\left(\frac{1\,\text{V}}{10\,\text{k}\Omega} + \frac{2\,\text{V}}{20\,\text{k}\Omega} + \frac{-0.5\,\text{V}}{10\,\text{k}\Omega}\right) = -(1 + 1 - 0.5)\,\text{V} = -1.50\,\text{V}$. The inverting input is a virtual ground: $V_- = 0\,\text{V}$.

b) $I = 0.1\,\text{mA} + 0.1\,\text{mA} - 0.05\,\text{mA} = 0.15\,\text{mA}$, which gives $V_{\text{out}} = -10\,\text{k}\Omega \cdot 0.15\,\text{mA} = -1.5\,\text{V}$.

c) Weight $-R_f/R_3 = -0.25 \Rightarrow R_3 = 40\,\text{k}\Omega$ (e.g. $39\,\text{k}\Omega + 1\,\text{k}\Omega$).

d) ▶ Falstad: `examples/opamps/opamp_summing.txt`

`Vout` $= -1.50\,\text{V}$, `V_sum_node` $= 0.00\,\text{V}$.

Exercise 6.3: Difference amplifier and resistor matching

A difference amplifier with $R_1 = R_3 = 10\,\text{k}\Omega$ and $R_2 = R_4 = 100\,\text{k}\Omega$ amplifies a bridge signal: $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$.

a) Calculate differential and common-mode voltage and the output voltage.
b) R_4 is $1\,\%$ too high ($101\,\text{k}\Omega$). Calculate the output voltage.
c) Calculate the common-mode gain and the CMRR for b).
d) Estimate the worst-case CMRR with $1\,\%$ resistors.
e) Check a) and b) in Falstad.

Solution 6.3

a) $V_D = 5\,\text{mV}$, $V_{CM} = 2.5025\,\text{V}$; $V_{\text{out}} = 10 \cdot 5\,\text{mV} = 50.0\,\text{mV}$.

b) $V_+ = 2.505\,\text{V} \cdot \frac{101}{111} = 2.279\,32\,\text{V}$, $V_{\text{out}} = V_+\,(1+10) - 2.5\,\text{V} \cdot 10 = 25.0726\,\text{V} - 25\,\text{V} = 72.6\,\text{mV}$: $45\,\%$ too high.

c) $A_{CM} = 11 \cdot \frac{101}{111} - 10 = 0.0090$ (input both at $2.5\,\text{V}$ gives $22.5\,\text{mV}$); $A_D \approx 10$; $\text{CMRR} = 10/0.009 \approx 1100$ ($61\,\text{dB}$).

d) $\text{CMRR} \approx (1+G)/(4\varepsilon) = 11/0.04 = 275$ ($49\,\text{dB}$): with $2.5\,\text{V}$ common mode up to $9\,\text{mV}$ error at the input, i.e. almost half of the full bridge span. Matched resistors or an integrated difference amplifier are needed.

e) ▶ Falstad: `examples/opamps/opamp_diffamp.txt`

`Vout_matched` $= 50.0\,\text{mV}$, `Vout_mismatch` $= 72.6\,\text{mV}$.

Exercise 6.4: Finite open-loop gain

A signal of 1 mV DC is amplified with non-inverting stages ($R_1 = 1\,\text{k}\Omega$). The op-amps have $A = 10^5$ (as in Falstad).

a) Calculate V_{out} for one stage with $G = 1000$ and one stage with $G = 10000$. Give R_2 and the errors.

b) Calculate V_{out} for $G = 10 \cdot 100$ and $G = 100 \cdot 100$ in two stages.

c) How large is the error of the $G = 100$ stage at 1 kHz if the op-amp has GBW $= 1$ MHz?

d) Check a) and b) in Falstad.

Solution 6.4

a) $G = 1000$: $R_2 = 999\,\text{k}\Omega$, $G_\text{real} = 1000/(1+0.01) = 990.1$, $V_\text{out} = 0.990\,\text{V}$ ($-1\,\%$). $G = 10000$: $R_2 = 9.999\,\text{M}\Omega$, $G_\text{real} = 10000/1.1 = 9091$, $V_\text{out} = 9.09\,\text{V}$ ($-9.1\,\%$).

b) $10/(1+10^{-4})\cdot 100/(1+10^{-3}) = 9.999\cdot 99.90 = 998.9$: $V_\text{out} = 0.999\,\text{V}$ ($-0.11\,\%$). $(100/1.001)^2 = 9980$: $V_\text{out} = 9.98\,\text{V}$ ($-0.2\,\%$).

c) $A(1\,\text{kHz}) = 1\,\text{MHz}/1\,\text{kHz} = 1000$, error $\approx G/A = 10\,\%$ (and a phase shift). Falstad does not show this, because it does not simulate the GBW.

d) ▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

`Vout_1000` $= 0.990\,\text{V}$, `Vout_10x100` $= 0.999\,\text{V}$,
`Vout_10000` $= 9.09\,\text{V}$, `Vout_100x100` $= 9.98\,\text{V}$.

Exercise 6.5: Offset voltage and bias current

A non-inverting amplifier has $R_1 = 1\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$; its input is at $0\,\text{V}$.

a) The op-amp has an offset voltage of $1\,\text{mV}$. Calculate V_{out}.

b) The input is connected to ground through $R_S = 100\,\text{k}\Omega$ and a bias current of $100\,\text{nA}$ flows into the non-inverting input (no offset). Calculate V_{out}.

c) The capstone amplifier has $G = 107.6$ and the op-amps have $2\,\text{mV}$ offset (referred to the input). Compare the output error with the full scale of $4\,\text{V}$ and with one LSB of the 3-bit ADC.

d) Check a) and b) in Falstad.

Solution 6.5

a) $V_{\text{out}} = (1 + 100) \cdot 1\,\text{mV} = 0.101\,\text{V}$.

b) $V_+ = -R_S I_B = -100\,\text{k}\Omega \cdot 100\,\text{nA} = -10\,\text{mV}$, $V_{\text{out}} = 101 \cdot (-10\,\text{mV}) = -1.01\,\text{V}$. A CMOS op-amp with $I_B = 1\,\text{pA}$ would give only $10\,\mu\text{V}$.

c) $107.6 \cdot 2\,\text{mV} = 0.215\,\text{V}$: $5.4\,\%$ of the full scale, 0.43 LSB (LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$). The zero must be checked and trimmed, or a zero-drift op-amp used.

d) ▶ Falstad: `examples/opamps/opamp_offset.txt`

`Vout_offset` $= 0.101\,\text{V}$, `V_plus_b` $= -10.0\,\text{mV}$, `Vout_bias` $= -1.01\,\text{V}$.

Exercise 6.6: Capstone: instrumentation amplifier with 4.00 V full scale

The Pt100 of the capstone ($100\,\Omega$ at $0\,°C$, $138.5\,\Omega$ at $100\,°C$) sits in a bridge fed from the 4.00 V reference: R_1 ($3.9\,k\Omega$) above the Pt100 gives V_S, $R_3 = 3.9\,k\Omega$ above $R_4 = 100\,\Omega$ gives V_R. The amplifier consists of two followers, a difference amplifier and a non-inverting stage. Its output shall be 0 V at $0\,°C$ and exactly 4.00 V at $100\,°C$.

a) Calculate V_R, V_S and the bridge output $V_{BR} = V_S - V_R$ at $0\,°C$ and $100\,°C$. What is the common-mode voltage?

b) Calculate the total gain. Why is it split into two stages?

c) Choose $G_1 = 10$ for the difference amplifier and give its resistors.

d) Dimension the second stage with $R_g = 10\,k\Omega$. Give the exact R_F, the nearest E96 value and a version with $91\,k\Omega$ plus a $10\,k\Omega$ trimmer (setting and full-scale range).

e) The three op-amp INA could replace the followers. Which R_G gives an input stage gain of 10 with $R = 10\,k\Omega$?

f) A 50 Hz interference of 5 mV amplitude is in series with the Pt100. How large is it at the output? What follows for the supply voltage and the next stage?

g) Check the design in Falstad.

Solution 6.6

a) $V_R = 4\,\text{V} \cdot 100/4000 = 0.100\,\text{V}$. $0\,°\text{C}$: $V_S = 0.100\,\text{V}$, $V_{BR} = 0$ (sensor current $1.000\,\text{mA}$). $100\,°\text{C}$: $V_S = 4\,\text{V} \cdot 138.5/4038.5 = 0.137\,18\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$. The common mode is only about $0.1\,\text{V}$: the op-amp inputs must work close to the negative rail.

b) $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$. One stage would be possible (error $G/A = 0.11\,\%$), but two stages of about 10 keep each error at $0.01\,\%$ and the bandwidth high, follow the lab rule $G < 100$, and separate zero (bridge) and span (R_F).

c) $R_1 = R_3 = 10\,\text{k}\Omega$, $R_2 = R_4 = 100\,\text{k}\Omega$ ($0.1\,\%$ or matched). $V_{\text{diff}} = 10 \cdot 37.18\,\text{mV} = 371.8\,\text{mV}$ at $100\,°\text{C}$.

d) $G_2 = 4.00\,\text{V}/371.8\,\text{mV} = 10.758$, $R_F = 9.758 \cdot 10\,\text{k}\Omega = 97.58\,\text{k}\Omega$. E96: $97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$. Trimmer version: $91\,\text{k}\Omega$ + $R_{\text{trim}} = 6.6\,\text{k}\Omega$; range $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V}$ … $4.13\,\text{V}$.

e) $1 + 2R/R_G = 10 \Rightarrow R_G = 2R/9 = 2.22\,\text{k}\Omega$; then the difference stage can have gain 1 and the second stage stays as it is.

f) At the tap: $5\,\text{mV} \cdot \frac{3.9\,\text{k}\Omega}{3.9\,\text{k}\Omega + 100\,\Omega} = 4.875\,\text{mV}$, at the output $\times 107.6 = 0.52\,\text{V}$ amplitude: the output swings up to $4.52\,\text{V}$. The op-amps need headroom (e.g. $\pm 15\,\text{V}$ as in Falstad); the low-pass of chapter 7 must attenuate the interference below $\text{LSB}/2 = 0.25\,\text{V}$.

g) ▶ Falstad: `examples/opamps/opamp_ina.txt`
with the Pt100 at $138.5\,\Omega$ the labels show `VB_minus` $= 0.100\,\text{V}$, `V_diff` $= 372\,\text{mV}$ and `Vout` $= 4.00\,\text{V}$; with the slider at $100\,\Omega$ `Vout` $= 0.00\,\text{V}$.